

IMPRS UFAST focus course
**High harmonic generation and ultrafast
electron dynamics simulations**
Alba de las Heras

Abstract:

Strong-field physics investigates the interaction of matter with high-intensity laser pulses whose electric field is comparable to, or exceeds, the Coulombic potential binding valence electrons. In this highly nonperturbative regime of nonlinear optics, the electronic potential is significantly distorted by the external field. Ultrafast electronic processes like tunnel ionization and high-order harmonic generation become the dominant dynamics. As a coherent, table-top source of extreme ultraviolet and soft X-ray radiation, high harmonic generation enables the synthesis of attosecond pulses and constitutes a paramount tool for high-resolution spectroscopy and imaging.

The course aims to introduce the fundamental concepts and theoretical modelling of strong-field ionization, high harmonic generation, and attosecond science. While we will mainly focus on laser-atom interactions, the last session will address the peculiarities of solid-state targets.

Topics include:

- Fundamentals of strong-field physics
- Historical milestones and current perspectives in attosecond science
- Single-electron and multi-electron methods for modelling strong-field phenomena or pump-probe schemes
- Basic time-dependent calculations of laser-matter interactions with Octopus software
- Macroscopic picture of high harmonic generation and control of the properties of structured attosecond pulses

Bldg. 900 (MPSD), SR O1.020

17th, 18th, 24th, 25th August 2026

09:00 to 12:00 h

Register on Geventis: I-UF FC10

Registration deadline:
10th August 2026

